

round4_GO

BOB-169 — export charset corruption: review GO (round 4)

Revision: 1 Last modified: 2026-08-25T17:50:00Z

Four review rounds on the §11.4.209 substrate, iterated to zero findings per §11.4.134. **BOB-169 is NOT closable on this change** — see “Still owed” below.

What was wrong

Three producers emitted HTML with no `<meta charset>`; weasyprint rendered PDFs from those body fragments, fell back to a non-UTF-8 codec, and baked mojibake into the PDF **text layer** — not a display artifact, corruption in the bytes.

docs/CONTINUATION.pdf — the §12.10 session-resumption document — carried **336 corrupted lines** and regenerated corrupt on every run. Corpus-wide, **301 of 334** exports carried no charset declaration while every export-sync gate reported green.

The three producers, now in lockstep

Producer	Fix
scripts/ generate_markdown_exports.sh	--standalone + charset-aware staleness + PDF staleness keyed on the HTML
scripts/regenerate-continuation-exports.sh	--standalone (the third producer, found only because the corpus scan was run)
docs_chain derived.go:139	already passed --standalone

Verified closed, by round

Finding	Closure
R2-F1 (BLOCKING) — PDF staleness ignored the HTML, so a fixed HTML never re-rendered its PDF	<pre>[[! -f "\$pdf" \ \ "\$md" -nt "\$pdf" \ \ "\$html" -nt "\$pdf"]].</pre> Reviewer re-ran the real-corpus shape on an unmodified copy: BEFORE <code>charset=0 pdf_moj=1</code> → AFTER <code>charset=1 pdf_moj=0</code> , plus an idempotence control it added and I had not run: a second consecutive run leaves the PDF md5 unchanged, so the clause heals without churning clean pairs
R2-F2 — gate renumbering	Re-derived a third time on a three-shape needed <code>instrument: sites=50</code> <code>{bracket:41, run_const_gate:9},</code> <code>denominators [50], 1..50</code> complete, zero duplicates, strictly increasing
R2-F3 — baseline honesty	Header states BASELINE is a CONSTANT, nothing lowers it, tightening is manual
R2-F5 / R3-F1 — my inserted paragraph severed a sentence	Reflowed; sentence reads whole; <code>bash -n clean</code>

Round-4 regression sweep (reviewer-executed)

<code>test_export_pdf_charset_integrity.sh</code>	2 passed, 0 failed
<code>test_cm_export_charset_valid.sh</code>	6 passed, 0 failed
CM-EXPORT-CHARSET-VALID (invariant 50)	PASS at baseline 301, no regression
docs/CONTINUATION	charset 1 , mojibake 0 (was 336)
§11.4.263 / residue / staging	clean; HEAD still dd9fcb5

The comment-only claim — two needles, not one

I proved the reflow was comment-only by stripping full-line comments and blanks from HEAD's copy and the working copy and diffing: 26 body lines each, exactly one differing (pandoc ... -o → pandoc ... --standalone -o, the fix itself). I ran ONE needle (echo NEEDLE → diff fires).

The reviewer ran **two**, and the second is the better instrument: it appended a comment-only line and confirmed the body diff stays SILENT. One positive needle proves an instrument is not dead; it does not prove it is aimed correctly. With both, "comment-only" is a measured property rather than an inference.

Still owed — stated so nothing reads as closed

Acceptance (c) — bulk regeneration of the 301 charset-less exports. Mechanism present, correctly sequenced behind R2-F1, **run not performed**, needs a quiescent tree.

BOB-182 — the §11.4.224(E) ratchet-adoption decision. Filing it satisfies the TRACKING obligation, not the DECISION obligation: the clause requires the operator's answer recorded as consumer DATA *before first enforcement*, and invariant 50 is already enforcing. The item stays OPEN. The interim is deliberate and matches §11.4.201(8)'s own precedence for the §11.4.135 ratchet — pinned at baseline it can only fire on regression, so enforcement today is near-inert while the decision is owed.

Carrier/blindness census for this item — five instances, three of them mine

1. `grep -qi 'charset'` in my own test matched charset-fixture and PASSED against a broken generator
2. my bracket-only renumber audit saw 41 of 50 labels
3. the reviewer's bracket-only scan, same blindness
4. the reviewer's name-requiring scan, which silently dropped invariants 1-4 and 11-15
5. mojibake counted inside quoted examples

Every one was caught by the same discipline: **refuse to report a zero until a needle of the same shape as the query proves the instrument can see it.**

What could not be analysed (reviewer's own statement, carried verbatim in substance)

- The full 50-invariant sweep — invariant 50 run in isolation, numbering audited statically; the other 49 gates' behaviour under the renumbered file unverified.
- The nine `run_const_gate` gate scripts themselves — only their labels.
- weasyprint's specific fallback codec — established non-UTF-8 and charset-driven, not identified.
- The python-markdown fallback branch — never executes on this host, unexercised by the guard.
- Rendering fidelity beyond the text layer.
- The tree was never quiescent across any round.